

COMPUTER NETWORKS

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5CS15	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES

- To introduce the fundamentals of various types of computer networks
- To demonstrate the TCP/IP and OSI models with merits and demerits
- To explore the various layers of OSI model
- To introduce UDP and TCP models

COURSE OUTCOMES

- Analyze computer networks and its components.
- Analyze the different types of network topologies and protocols.
- Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.
- Select and use various sub netting and routing mechanisms.
- Design a network diagram for different use cases.

UNIT-I	INTRODUCTION	CLASSES: 14
INTRODUCTION: Network applications, network hardware, network software, reference models: OSI, TCP/IP, Internet, Connection oriented network - X.25, frame relay. THE PHYSICAL LAYER: Theoretical basis for communication, guided transmission media, wireless transmission, the public switched telephone networks, mobile telephone system.		
UNIT-II	THE DATA LINK LAYER	CLASSES: 14
THE DATA LINK LAYER: Design issues, error detection and correction, elementary data link protocols, sliding window protocols. THE MEDIUM ACCESS SUBLAYER: Channel allocations problem, multiple access protocols, Ethernet, Data Link Layer switching, Wireless LAN, Broadband Wireless, Bluetooth		
UNIT-III	THE NETWORK LAYER	CLASSES: 10
THE NETWORK LAYER: Network layer design issues, routing algorithms, Congestion control algorithms, Internet working, the network layer in the internet (IPv4 and IPv6), Quality of Service.		
UNIT-IV	THE TRANSPORT LAYER	CLASSES: 10
THE TRANSPORT LAYER: Transport service, elements of transport protocol, Simple Transport Protocol, Internet transport layer protocols: UDP and TCP.		
UNIT-V	THE APPLICATION LAYER	CLASSES: 12
THE APPLICATION LAYER: Domain name system, electronic mail, World Wide Web: architectural overview, dynamic web document and http. APPLICATION LAYER PROTOCOLS: Simple Network Management Protocol, File Transfer Protocol, Simple Mail Transfer Protocol, Telnet.		
TEXT BOOKS - Computer networks-Andrew S Tanenbaum, 5th edition, pearson education - Data communication and networking-Behrouz. A. Forouzan , fifth edition, TMH, 2013		

REFERENCE BOOKS

1. Behrouz A. Forouzan (2006), Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Pearson Education, India.

WEB LINKS

1. <https://nptel.ac.in/courses/106/105/106105183/>
2. <https://freecomputerbooks.com/networkComputerBooks.html>